Symbolic Tumor-Growth ODE Candidate Ranking

Independent blinded evaluation. Current date/time: 2026-09-20 23:34 (UTC–05:00). Trial information: Gaussian noise 5.0%; training interval [0.0, 15.0]; forecast interval [15.5, 25.5].

Rank	Candidate ID	Main strength	Main concern	Reason for rank
1	C9	Simple decreasing quadratic with positive growth at low V and stable saturation; no restricted-domain singularity.	Training MSE is not the absolute minimum, but it is low while complexity remains moderate.	Best balance of robustness, low training MSE (0.792102), and moderate complexity (11). Complexity = 11; training trajectory MSE = 0.792102.
2	C7	Simple decreasing quadratic with positive low-volume growth and stable saturation; no singularity.	Slightly higher training MSE than C9.	Shares C9's simple robust structure, but its training MSE (1.049418) is higher. Complexity = 11; training trajectory MSE = 1.049418.
3	C13	Explicit saturation factor gives positive growth below the listed equilibrium and negative growth above it.	Higher complexity than the quadratic candidates and a slightly higher MSE than C9.	Its explicit saturation structure is robust; the MSE (1.053478) is close to C7, with complexity 14. Complexity = 14; training trajectory MSE = 1.053478.
4	C21	Monotone fifth-power decline gives positive low-volume growth and a stable saturation without a finite-domain singularity.	Higher complexity and slightly higher MSE than the leading low-complexity candidates.	Robust globally bounded structure; MSE 1.058234 is close to the preceding candidates, but complexity is 18. Complexity = 18; training trajectory MSE = 1.058234.
5	C23	Same robust monotone fifth-power structure as C21, with better training MSE.	Higher complexity than C21 despite its lower training MSE.	Very robust structure and the lower MSE of the fifth-power pair (1.021296), but complexity is 18. Complexity = 18; training trajectory MSE = 1.021296.
6	C20	Positive low-volume growth with a negative highest-order term, giving a bounded saturation structure.	More complex than the simpler quadratic/logistic-like candidates.	Bounded polynomial behavior with MSE 0.980523; complexity 17 places it behind simpler robust forms. Complexity = 17; training trajectory MSE = 0.980523.
7	C18	Positive low-volume growth and negative highest-order term provide bounded large-V behavior.	Higher complexity and a slightly higher MSE than the strongest lower-complexity candidates.	Bounded polynomial behavior with MSE 1.044230; complexity 17 limits its parsimony. Complexity = 17; training trajectory MSE = 1.044230.
8	C15	Low-to-moderate complexity polynomial with positive low-volume growth and eventual negative growth.	More complex than C9/C7 and with a somewhat higher MSE.	Reasonable bounded polynomial behavior and MSE 1.046150, but complexity 16. Complexity = 16; training trajectory MSE = 1.046150.
9	C24	Bounded high-volume behavior from its negative highest-order term and positive low-volume RHS.	Higher complexity and slightly higher MSE than C15/C18.	Bounded high-volume behavior, but complexity 19 and MSE 1.079208 are less favorable. Complexity = 19; training trajectory MSE = 1.079208.
10	C22	Monotone fifth-power form with stable saturation and no finite-domain singularity.	Training MSE is relatively high for its complexity.	Very robust monotone structure, but its MSE 2.021023 is substantially worse. Complexity = 18; training trajectory MSE = 2.021023.
11	C16	Negative highest-order term provides bounded large-V behavior despite the higher-order polynomial.	Training MSE is high despite its bounded high-volume behavior.	Bounded high-volume behavior, but MSE 2.100447 is poor relative to alternatives. Complexity = 16; training trajectory MSE = 2.100447.
12	C8	Simple quadratic saturation structure with no singularity and positive low-volume RHS.	Training MSE is comparatively high.	Simple robust quadratic structure, but MSE 2.083470 is substantially worse. Complexity = 11; training trajectory MSE = 2.083470.
13	C14	Has an explicit saturation factor and no logarithmic domain restriction.	At very small positive V its RHS is negative, which is less consistent with sustained positive tumor growth.	Despite good MSE (0.932772), its negative low-volume RHS is a major biological-robustness concern, so it is placed below candidates with positive low-volume growth. Complexity = 14; training trajectory MSE = 0.932772.

Rank	Candidate ID	Main strength	Main concern	Reason for rank
14	C3	Very low complexity and a logarithmic saturation form with a finite positive equilibrium.	The logarithm's domain boundary lies not far above the saturation region, creating extrapolation fragility.	Low complexity helps, but the logarithmic domain restriction is a major robustness concern; MSE 1.521070. Complexity = 6; training trajectory MSE = 1.521070.
15	C2	Very low complexity and a logarithmic saturation form with a finite positive equilibrium.	The logarithm's domain boundary lies not far above the saturation region, creating extrapolation fragility.	Low complexity and similar structure to C3, but slightly worse MSE (1.523209) with the same domain concern. Complexity = 6; training trajectory MSE = 1.523209.
16	C1	Very low complexity and a logarithmic saturation form with a finite positive equilibrium.	The logarithm's domain boundary lies not far above the saturation region, creating extrapolation fragility.	Low complexity, but MSE 1.836632 is worse than C2/C3 and the log boundary remains a concern. Complexity = 6; training trajectory MSE = 1.836632.
17	C6	Low complexity and a logarithmic saturation form with a finite positive equilibrium.	The logarithmic domain boundary is especially close to the saturation region.	MSE 1.203126 is better than the simpler log forms, but its log-domain boundary is closer to saturation. Complexity = 8; training trajectory MSE = 1.203126.
18	C4	Low complexity and a logarithmic saturation form with a finite positive equilibrium.	The logarithmic domain boundary is especially close to the saturation region.	MSE 1.230082, but the log-domain boundary is close to the saturation region. Complexity = 8; training trajectory MSE = 1.230082.
19	C5	Low complexity and a logarithmic saturation form with a finite positive equilibrium.	The logarithmic domain boundary is relatively close to the saturation region.	MSE 1.672765 and a nearby log-domain boundary make it less defensible. Complexity = 8; training trajectory MSE = 1.672765.
20	C10	Log-log form can represent saturation, but introduces an additional domain restriction.	Nested logarithms impose a more restricted domain and greater mathematical fragility.	MSE 1.159686 is moderate, but nested-log domain restrictions make robustness weaker. Complexity = 12; training trajectory MSE = 1.159686.
21	C12	Log-log form can represent saturation, but introduces an additional domain restriction.	Nested logarithms impose a more restricted domain and greater mathematical fragility.	MSE 1.123532 is somewhat better than C10, but the same nested-log fragility remains. Complexity = 12; training trajectory MSE = 1.123532.
22	C11	Log-log form has an additional domain restriction and comparatively high training MSE.	Nested logarithms impose a more restricted domain, and the MSE is high.	Nested-log fragility plus MSE 2.021433 makes it difficult to justify over the robust polynomial alternatives. Complexity = 12; training trajectory MSE = 2.021433.
23	C17	Excellent training MSE, but its positive highest-order term permits eventual growth rather than globally bounded saturation.	The leading positive high-order term creates an eventual unbounded-growth regime, weakening extrapolation robustness.	The lowest MSE (0.753099) cannot compensate for the eventual positive high-order growth term under the robustness-first criterion. Complexity = 16; training trajectory MSE = 0.753099.
24	C26	Excellent training MSE, but its positive highest-order term permits eventual growth rather than globally bounded saturation.	The leading positive high-order term creates an eventual unbounded-growth regime, weakening extrapolation robustness.	The second-lowest MSE (0.779387) is outweighed by the eventual positive high-order growth term and higher complexity. Complexity = 19; training trajectory MSE = 0.779387.
25	C19	Negative highest-order term supports bounded behavior, but the training MSE is substantially worse than simpler robust candidates.	Although bounded at high V, its MSE is much worse than several simpler bounded candidates.	Its bounded high-V term is favorable, but MSE 1.754674 is poor relative to other bounded candidates. Complexity = 17; training trajectory MSE = 1.754674.
26	C25	Negative highest-order term supports bounded behavior, but the training MSE is substantially worse than simpler robust candidates.	Although bounded at high V, its MSE is high and complexity is also high.	Its bounded high-V term is favorable, but MSE 1.930357 and complexity 19 are both unfavorable. Complexity = 19; training trajectory MSE = 1.930357.

Top choice: C9

C9 combines a simple quadratic saturation structure with positive low-volume growth, stable saturation, and no evident finite-domain singularity. Its training trajectory MSE of 0.792102 is substantially lower than the other candidates with similarly modest complexity, while the robustness-first criterion keeps the lower-MSE high-order candidates below it.

Across the candidate set, the most important distinction is between bounded, nonsingular growth structures and expressions whose extrapolation is limited by nearby logarithmic domain boundaries or eventual positive high-order terms. Within the more robust group, lower training MSE generally separates candidates, followed by complexity when the fit is otherwise close.